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NexusIndex: A Self-Optimizing Multimodal Framework for Fake News Detection with Dynamic Indexing and Retrieval

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Introduction: NexusIndex for Multimodal Fake News Detection

- **Motivation:** Prior IR systems lack the capability to jointly analyze text and images, limiting their effectiveness in identifying multimodal misinformation. NexusIndex addresses this by aligning visual and textual semantics through unified embeddings.
- **Challenges:** Existing pre-trained language models (PLMs) and Bi-encoder techniques enhance efficiency through offline indexing, but their use of single-vector representations limits their ability to capture fine-grained semantic relationships, reducing effectiveness compared to cross-encoders.
- **Proposed Solution:** NexusIndex introduces a two-stage framework that integrates Transformer-based text embeddings and MobileNetV3 visual features, enhanced with attention and a **FAISSNexusIndex layer** for fast similarity retrieval and semi-supervised learning to handle evolving misinformation.

Methods: NexusIndex Framework Overview

- **Cross-Modal Embedding Fusion**

NexusIndex uniquely integrates Transformer-based text embeddings and MobileNetV3 image features, fused via cross-attention to capture rich multimodal semantics.

- **FAISSNexusIndex Layer (Novel Contribution)**

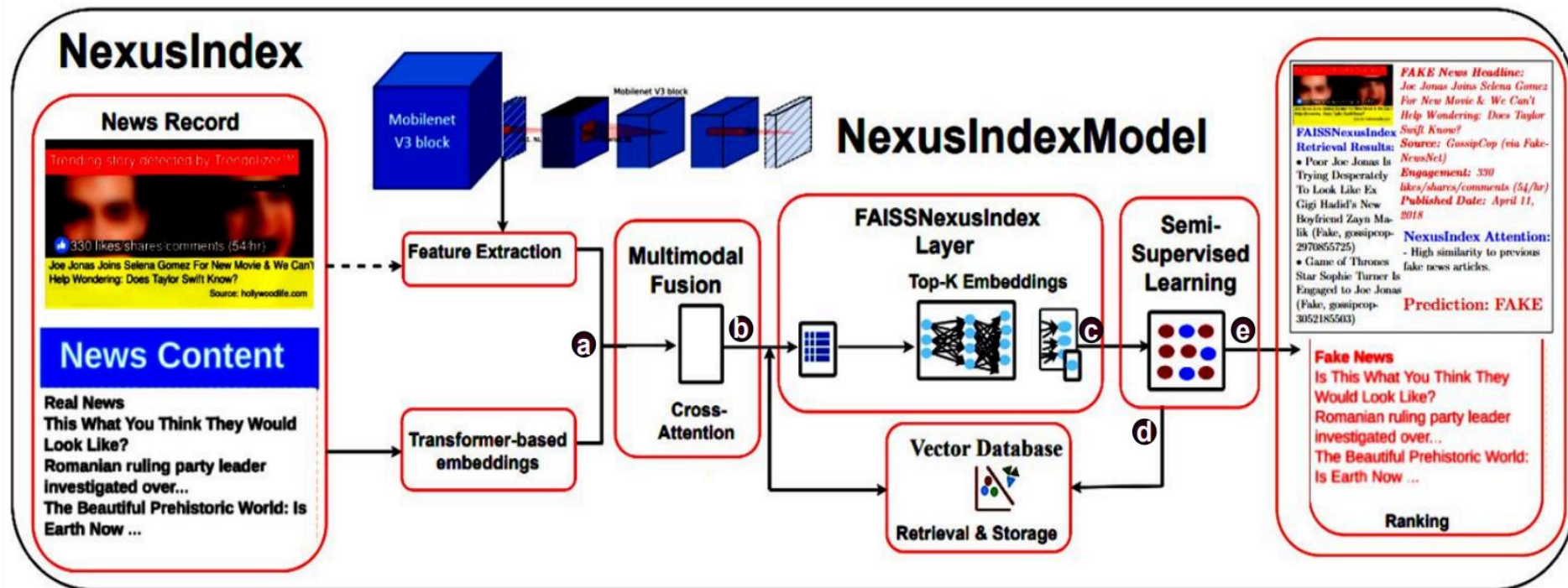
Introduced as a custom neural layer, it enables fast, in-network top-k similarity retrieval, optimizing both classification and retrieval within a unified framework.

- **Dynamic Semi-Supervised Learning (Novel Contribution)**

Integrates pseudo-labeling with Local Variance Filtering (LVF) to adapt to evolving misinformation using both labeled and unlabeled data.

NexusIndex System Architecture

This diagram illustrates the end-to-end workflow of the NexusIndex framework



- (a) Feature Extraction (b) Multimodal Fusion (c) FAISSNexusIndex Layer (Novel Contribution)
(d) Semi-Supervised Learning & Vector Retrieval (Novel Contribution) (e) Final Classification and Ranking

Algorithms Overview — NexusIndex Models

- Algorithm 1: NexusIndexModel I

Text and image embeddings with *fixed weights* W_t, W_i

FAISS for efficient similarity-based retrieval

3-layer neural network is trained to classify news as real or fake

Learns model parameters θ via supervised loss

- Algorithm 2: NexusIndexModel II (Enhanced)

Learns **dynamic weights** via attention mechanism

$$\tilde{e}_x = \text{softmax}(W_a e_x) \odot e_x$$

Integrates **FAISSNexusIndex Layer** for top-k retrieval

cross-modal attention for better semantic alignment

Jointly learns attention weights W_a and model parameters θ

Updates parameters with gradient descent $\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$

Algorithm 1: Multimodal NexusIndexModel I

Input: Real News R , Fake News F , Text Model M_{text} ,
Image Model M_{image}

Output: Trained NexusIndex Model, Accuracy

- 1 $E \leftarrow W_t \cdot M_{\text{text}}(R, F) + W_i \cdot M_{\text{image}}(R, F)$
 - 2 $I \leftarrow \text{FAISS}(\{E_R, E_F\})$
 - 3 $(X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}}) \leftarrow \text{Preprocess}(E)$
 - 4 Define neural network:

$$f_{\theta}(x) = \sigma(W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \cdot x + b_1) + b_2) + b_3)$$
 - 5 Train model: $\theta \leftarrow \arg \min_{\theta} \sum_{i=1}^N L(f_{\theta}(x_i), y_i)$
 - 6 Compute accuracy:

$$\text{Acc} \leftarrow \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{round}(f_{\theta}(x_i)) = y_i)$$
 - 7 **return** Trained model, Accuracy
-

Algorithm 2: Multimodal NexusIndexModel II

Input: Datasets R, F ; Models $M_{\text{text}}, M_{\text{image}}$;
Neighbors $k = 5$

Output: Trained model θ , Metrics

- 1 **foreach** $x \in R \cup F$ **do**
 - 2 $e_x \leftarrow W_t M_{\text{text}}(x_{\text{text}}) + W_i M_{\text{image}}(x_{\text{image}})$
 - 3 Assign labels: $y_r \leftarrow 1$ for $r \in R$, $y_f \leftarrow 0$ for $f \in F$
 - 4 $\tilde{e}_x \leftarrow \text{softmax}(W_a e_x) \odot e_x$;
 $(D_q, I_q) \leftarrow \text{FAISSNexusIndex}(\tilde{e}_x)$
 - 5 $\hat{y} \leftarrow \sigma(W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \tilde{e}_x + b_1) + b_2) + b_3)$
 - 6 $L(\theta) \leftarrow -\frac{1}{N} \sum [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$
 - 7 $\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$
 - 8 Compute metrics: Accuracy, Recall@ k , nDCG@ k
 - 9 **return** θ , Metrics
-

Algorithms Overview — Domain-Adaptive Pseudo-Labeling

Algorithm 3: RoBERTa Fine-Tuning for Pseudo-Labeling

Fine-tunes RoBERTa on labeled fake news data D_{fake} to adapt it to the fake news domain.
Fine-tuned model to assign pseudo-labels to *real news* samples D_{real} .
Output: Labeled D_{real} used for self-training in the semi-supervised module.

Algorithm 3: RoBERTa Fine-Tuning for Fake News Detection

Input: $D_{\text{real}}, D_{\text{fake}}$; RoBERTa model; Hyperparameters

Output: Labeled D_{real}

```
1 Initialize RoBERTa model:  $\theta$ 
2 foreach  $(x, y) \in D_{\text{fake}}$  do
3    $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(x, y, \theta)$  // Fine-tune with
   fake news
4 foreach  $x \in D_{\text{real}}$  do
5   Predict:  $\hat{y} \leftarrow \arg \max_{y'} P(y' | x, \text{RoBERTa}^*)$ 
6 return Labeled  $D_{\text{real}}$ 
```

STATISTICS FOR DIFFERENT DATASETS.

Dataset	Total	Fake	Real
Labeled ABCNews (Ours)	1,000	42	958
WELFake	72,134	37,106	35,028
A Million News	1,244,184	0	1,244,184
PolitiFact	21,152	11,760	9,392
GossipCop	22,140	5,323	16,817

Results: Pseudo-Labeling Performance

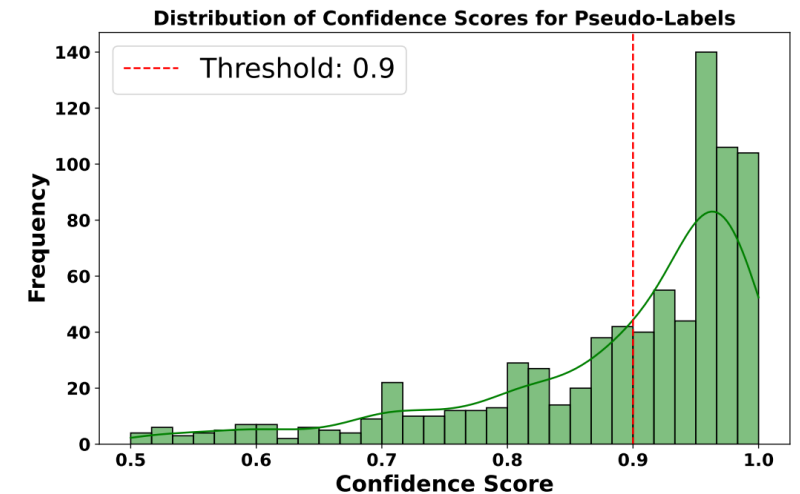
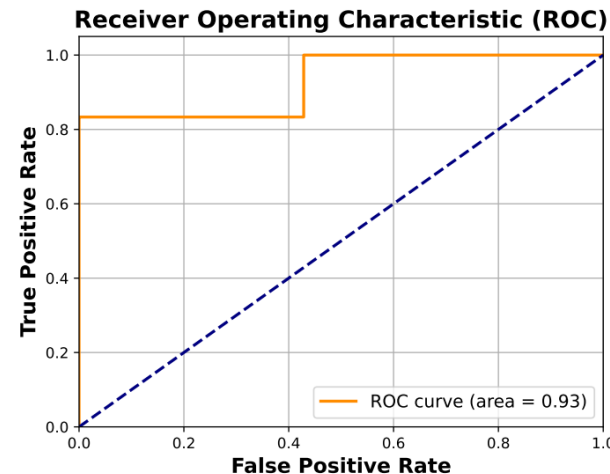
Technique: Applied pseudo-labeling + Local Variance Filtering (LVF) to ABCNews.

Confidence Filtering: Only pseudo-labels with confidence > 0.9 and low variance were retained.

Evaluation: ROC curve shows **AUC = 0.93**. High-confidence pseudo-labels improve model robustness

TABLE I
STATISTICS FOR DIFFERENT DATASETS.

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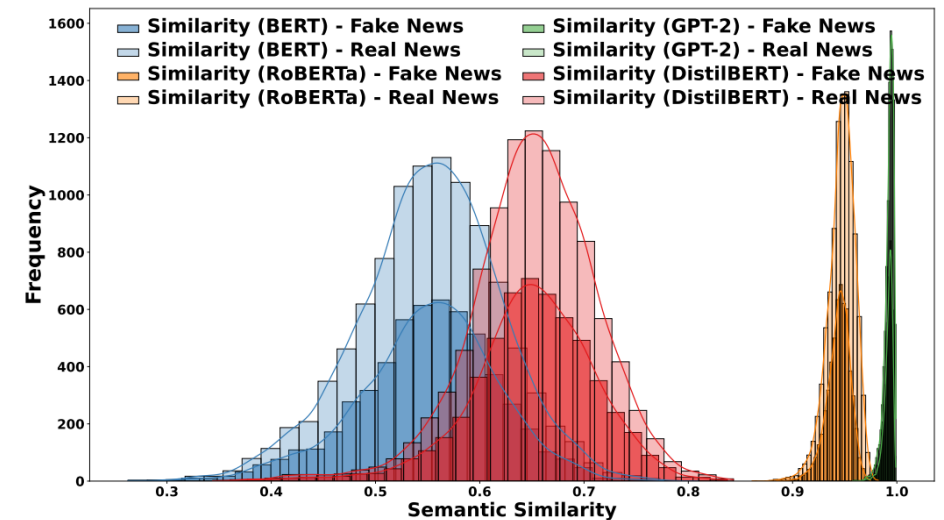
Results: Effect of Language Models on Semantic Similarity (Text Only)

Approach: Used BERT, RoBERTa, GPT, and DistilBERT embeddings to compute similarity scores between articles.

Findings:

- BERT and RoBERTa clearly separated real vs. fake news.
- GPT showed less variation.
- DistilBERT had a wider similarity score distribution.

Select the best language model embeddings for distinguishing real vs. fake news, improving overall system accuracy.



Results: Performance vs. Baselines – Ablation Study

Datasets: PolitiFact & GossipCop

F1-Score: PolitiFact: 0.9350 (best) GossipCop: 0.9115 (best)

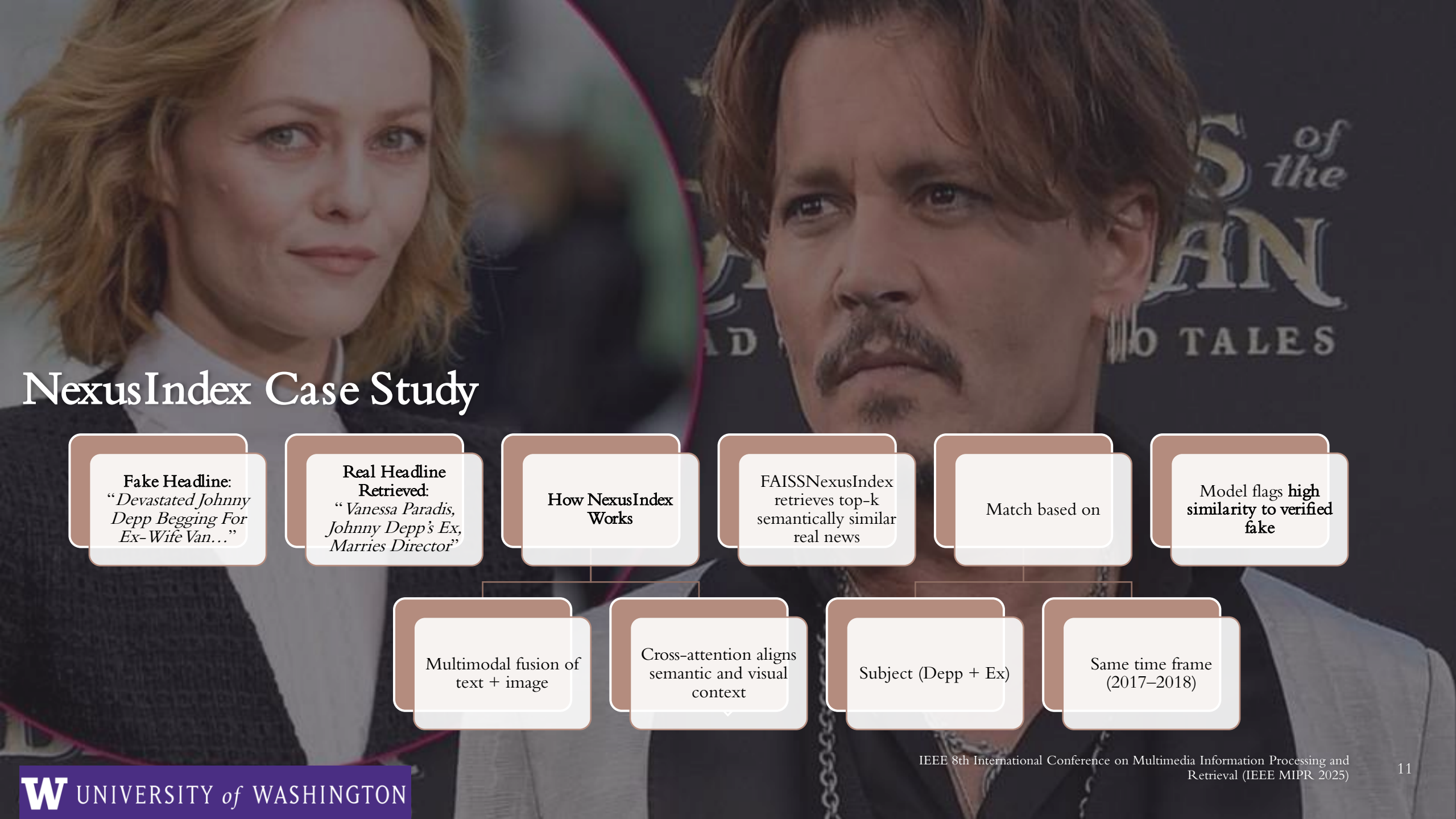
AUC: PolitiFact: 0.9530 GossipCop: 0.9205

Compared to top multimodal baseline (AKA-Fake): +0.54% F1 (PolitiFact) +0.28% F1 (GossipCop)

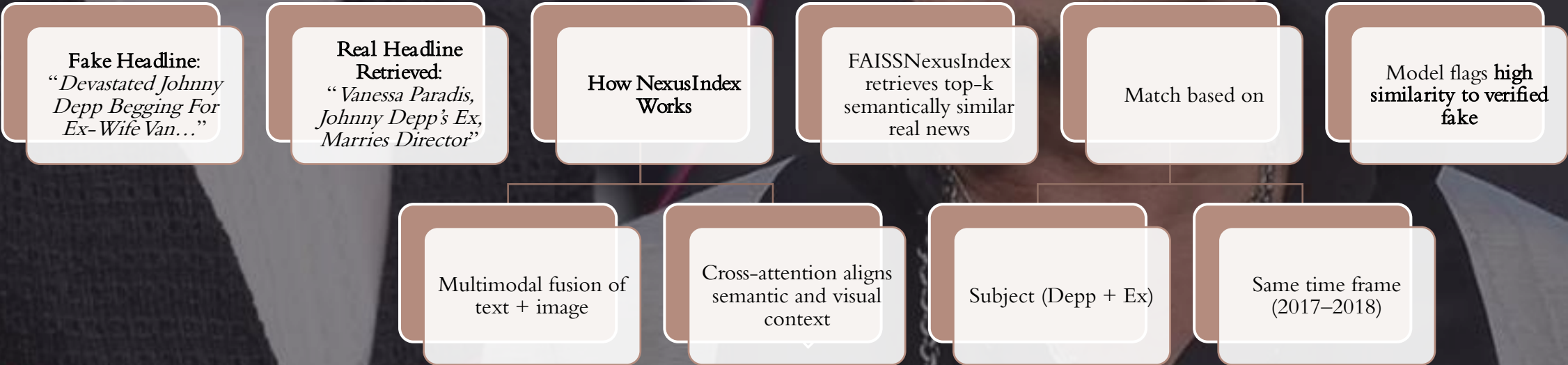
MUSER: Up to +28.19% F1 and +26.73% AUC on PolitiFact

Accuracy from 0.75 → 0.95 F1-Score from 0.71 → 0.91 (NexusIndexModel II – Novel Contribution)

Modality	Model	PolitiFact					GossipCop				
		Precision	Recall	F1-Score	Accuracy	AUC	Precision	Recall	F1-Score	Accuracy	AUC
Unimodal	SVM [38]	0.7460	0.6826	0.6466	0.6694	0.6826	0.7493	0.6254	0.5955	0.6643	0.6253
	RFC [16]	0.7470	0.7361	0.7362	0.7406	0.8074	0.7015	0.6707	0.6691	0.6918	0.7389
	DTC [39]	0.7476	0.7454	0.7450	0.7486	0.7454	0.6921	0.6922	0.6919	0.6959	0.6929
	GRU-2 [40]	0.7083	0.7048	0.7041	0.7109	0.7896	0.7176	0.7079	0.7073	0.7180	0.7516
	BERT [1]	0.619	0.599	0.608	0.608	0.608	0.664	0.649	0.635	0.635	0.635
	TextRNN [10]	0.641	0.579	0.594	0.615	0.564	0.671	0.646	0.590	0.615	0.576
Graph-based	GET [13]	0.712	0.770	0.725	0.725	0.725	0.712	0.710	0.751	0.751	0.725
	MUSER [18]	0.760	0.738	0.729	0.757	0.752	0.843	0.734	0.776	0.784	0.830
	FinerFact [12]	0.9196	0.9037	0.9172	0.9092	0.9384	0.8615	0.8779	0.8685	0.8320	0.8637
Knowledge-based	B-TransE [15]	0.7739	0.7658	0.7641	0.7694	0.7995	0.7369	0.7333	0.7334	0.7394	0.7995
	KCNN [41]	0.7852	0.7824	0.7804	0.7827	0.8488	0.7483	0.7422	0.7433	0.7491	0.8125
	KAN [17]	0.8687	0.8499	0.8539	0.8586	0.9197	0.7764	0.7696	0.7711	0.7766	0.8435
Multimodal	AKA-Fake [3]	0.9300	0.9300	0.9300	0.9190	0.9500	0.8800	0.9410	0.9090	0.8580	0.9180
	NexusIndex (Ours)	0.9330	0.9340	0.9350	0.9215	0.9530	0.8840	0.9430	0.9115	0.8610	0.9205
(Impv.)	NexusIndex vs. AKA-Fake	+0.32%	+0.43%	+0.54%	+0.27%	+0.32%	+0.45%	+0.21%	+0.28%	+0.35%	+0.22%
(Impv.)	NexusIndex vs. MUSER	+22.76%	+26.55%	+28.19%	+21.73%	+26.73%	+4.86%	+28.46%	+17.47%	+9.82%	+10.91%
Ablation Study. All improvements are reported as absolute scores.											
Model	Components	Accuracy	Precision	Recall	F1-Score	R@100	nDCG@10				
MultiModelSimNet	FAISS ✗, FAISSNexusIndex ✗, Network ✓	0.7500	0.7500	0.6667	0.7059	N/A	N/A				
NexusIndexModel I	FAISS ✓, FAISSNexusIndex ✗, Network ✓	0.8500 (+0.10)	0.8000 (+0.05)	0.8889 (+0.22)	0.8421 (+0.14)	N/A	N/A				
NexusIndexModel II	FAISS ✗, FAISSNexusIndex ✓, Network ✓	0.9500 (+0.20)	1.0000 (+0.25)	0.8333 (+0.16)	0.9091 (+0.20)	1.0	0.207				



NexusIndex Case Study



Conclusion & Future Work

NexusIndex: Scalable, modular system for multimodal fake news detection.

Two models:

NexusIndexModel I: text + image embeddings using fixed weights (W_t , W_i), with a similarity-based retrieval layer and neural network classifier.

NexusIndexModel II: Enhances Model I with:

- Learned dynamic weights
- Cross-attention for cross-modal alignment
- **FAISSNexusIndex Layer** for top-k retrieval

Results:

- Accuracy: **95%**
- Recall@100: **100%**

Future Directions:

- Improve retrieval ranking (e.g., optimize **nDCG@10**).
- Extend the system for **deepfake detection**.

PERFORMANCE METRICS FOR FAKE NEWS.

Class	Precision	Recall	F1-Score
Fake News	0.93	1.00	0.97
Real News	1.00	0.83	0.91
Overall Accuracy		0.95	
Macro Avg	0.97	0.92	0.94
Weighted Avg	0.95	0.95	0.95